

# Title

# AI FitTrack API

Presented by: KALAIYARASI M  
College : C.M Annamai art's and science college for women  
Department : computer science

## Introduction

AI FitTrack API is a fitness management backend system.

It helps users track fitness-related information.

Uses APIs to handle user and fitness data.

AI can provide personalized fitness insights.

## Objectives

Track daily fitness activities.

Store user fitness information.

Provide personalized recommendations.

Monitor fitness progress.

Make fitness data easy to access through APIs.

## Key Features

- User registration and login.
- Fitness activity tracking.
- Workout record management.
- Progress monitoring.
- AI-based fitness suggestions.
- REST API support.

## Slide 5 – Technologies Used

## Technologies Used

Backend: Node.js

Framework: Express.js

Database: MongoDB

API: REST API

AI: AI-based recommendation system

Testing: Postman

## System Architecture

User → REST API → Express.js Server → AI Processing → MongoDB

User sends request.

API receives and processes the request.

Fitness data is stored in MongoDB.

AI analyzes relevant data.

Response is returned to the user.

## API Modules

User API – Registration and login

Profile API – Manage fitness profile

Workout API – Add and view workouts

Progress API – Track fitness progress

Recommendation API – Generate AI-based suggestions

Database

MongoDB Collections:

Users

Fitness Profiles

Workouts

Progress

Recommendations

MongoDB stores the fitness information and makes it available to the API.

## Advantages & Future Scope

### Advantages:

Easy fitness data management.

Personalized recommendations.

Fast API-based access.

Progress can be monitored regularly.

### Future Scope:

Mobile application integration.

Wearable device integration.

Advanced AI recommendations.

Fitness chatbot.

## Conclusion

AI FitTrack API provides a simple fitness tracking solution.

It combines REST APIs, MongoDB and AI.

It helps manage and analyze fitness information.

The system can be extended into a complete AI-powered fitness application.

Thank You  
KALAIYARASI M